Fantastic 4 Team – Kaggle Pokémon Challenge

Ivan Dalla Ragione 2009768 | Valentina Foschi 2005787 | Francesco Anatriello 1957447 | Annika Unmüßig 2046421

Abstract

*This report describes the methodology used by our team in the FDS: Pokémon Battles Prediction 2026 challenge[**?**]. Our three final submissions share a single feature pipeline and differ only in the model: two weighted ensembles, and a standalone CatBoost model. Most of our work went into feature engineering, and into a cross-validation protocol for model selection. We conclude by discussing why the private ranking of our submissions differs from the CV one, and where we may have overfitted.*

1. Evaluated Approaches

Feature Engineering & Selection We built a single feature pipeline and a single validation protocol, and applied them to three modelling strategies (two ensembles and a standalone CatBoost), so that our submissions see the same information and differ only in how it is modelled. Each battle becomes 137 numeric features computed on the 30 observed turns, mostly as P1 − P2 differences: **battle state and dynamics** (HP, status, KOs, boosts, field effects, sustain; 122), **pairwise duels** based on the Gen 2 damage formula (6), **P2's full team** when its revealed Pokémon identify a training team (6), and three **learned ratings** of team, species and battle-state strength, computed out-of-fold to avoid leakage (3). CatBoost additionally receives 14 identity columns (teams, active Pokémon, last moves) as categorical features.

Model Experimentation & Refinement The base learners are a **boosting trio** (XGBoost, LightGBM, HistGradientBoosting, five seeds each, averaged), **CatBoost** (five seeds, native categorical handling) and an L2-regularised **logistic regression** (LR, $C = 0.05$), a smooth linear view that improved log-loss in every CV split. Hyperparameters were tuned by random search on CV log-loss. We combined them into three submissions that are strong but structurally differen:

1. **Full blend:** 40% boosting trio (13.3% per booster), 40% CatBoost, 20% LR; our best CV;
2. **CatBoost only:** our most stable single model across CV splits;
3. **Trio + LR:** 75% boosting trio (25% per booster), 25% LR; the only one that does not depend on CatBoost.

Validation Strategy We use stratified 5-fold cross-validation, refitting the learned ratings inside every training fold so that the validation fold is never seen. Since a single split is noisy, every change was tested on 4 repeated CV splits with a rule fixed in advance: mean accuracy and log-loss both better, and accuracy better in at least 3 of 4 splits. Final choices were confirmed on 4 fresh splits. Model selection relied on CV, not on the public leaderboard.

2. Results (CV: 5-fold out-of-fold, seed 42)

Submission	CV	Public	Private
Full blend	**0.7003**	0.70333	0.68542
CatBoost only	0.6968	**0.70400**	0.68657
Trio + LR	0.6984	0.69866	**0.68771**

Leaderboard Shift and Possible Overfitting All three submissions lost 0.011–0.018 from public to private and their order changed: the full blend, best in CV, became our weakest, while trio + LR, last on the public leaderboard, became our best. As a team we moved from 3rd to 8th place. Part of this can be noise: the public score uses 1,500 test battles and the private one 3,500 (standard errors of about 0.012 and 0.008), our private scores are only 4–8 battles apart, places 5–8 are within 5 battles of each other, and several teams climbed 19–27 positions. Still, the pattern suggests some overfitting that our protocol did not catch:

- **CatBoost dropped most.** The two submissions containing it lost 0.017–0.018, against 0.011 for trio + LR. It is the only model using the identity columns, of very high cardinality (about 4,200 teams, 5,700 sets of alive Pokémon), which may generalise worse than the numeric features used by the trio.
- **Our CV was optimistic.** It overestimated private accuracy by 0.010–0.015, most for the full blend. Many decisions, each worth 0.001–0.003, were taken on the same 10,000 battles; even with a fixed rule, reusing the same data so often makes CV favour configurations that fit it best.

In hindsight, the simplest ensemble, without identity columns, was the most robust; an untouched hold-out set would have been more honest.